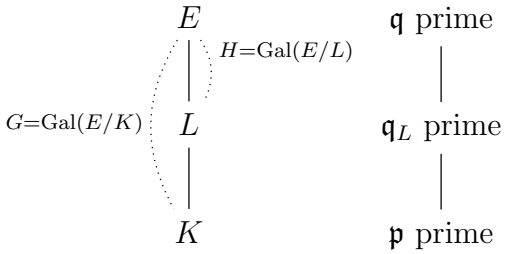
Algebraic Number Theory NMAL431
Exercise Sheet 3 - Solutions

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1. Let $G = \mathrm{Gal}(E/K)$. Let us note that, for any intermediate field L, where $K \subset L \subset E$, the extension E/L is also Galois, and we may thus define $H := \mathrm{Gal}(E/L)$. We also let $\mathfrak{q}_L := \mathfrak{q} \cap O_L$. We therefore have the following picture:



We note that the decomposition group $D_{\mathfrak{q}/\mathfrak{q}_L}$ satisfies

$$D_{\mathfrak{q}/\mathfrak{q}_L} := \{\sigma \in H \mid \sigma(\mathfrak{q}) = \mathfrak{q}\} = D_{\mathfrak{q}/\mathfrak{p}} \cap H$$

and the inertia group $I_{\mathfrak{q}/\mathfrak{q}_L}$ satisfies

$$I_{\mathfrak{q}/\mathfrak{q}_L} := \{\sigma \in H \mid \sigma(\mathfrak{q}) = \mathfrak{q} \text{ and } \sigma|_{O_E} \bmod \mathfrak{q} = \mathrm{id}\} = I_{\mathfrak{q}/\mathfrak{p}} \cap H$$

(a) In the case where $L = E^I$, then if some automorphism σ satisfies $\sigma \in I := I_{\mathfrak{q}/\mathfrak{p}}$, then by definition σ fixes E^I, and so $\sigma \in \mathrm{Gal}(E/E_I) = H$. Conversely, if $\sigma \in H$, then σ fixes E^I, and so $\sigma \in I = I_{\mathfrak{q}/\mathfrak{p}}$.[1]

Therefore $I_{\mathfrak{q}/\mathfrak{p}} = H$, which implies $I_{\mathfrak{q}/\mathfrak{q}_L} = I_{\mathfrak{q}/\mathfrak{p}} = H$. So in particular, $e_{\mathfrak{q}/\mathfrak{q}^I} = |I_{\mathfrak{q}/\mathfrak{q}^I}| = |H| = [E : E^I]$.

(b) We can do a similar argument as in part (a) in the case where $L = E^D$. Now we have that $D_{\mathfrak{q}/\mathfrak{p}} = H := \mathrm{Gal}(E/E^D)$, and thus $D_{\mathfrak{q}/\mathfrak{q}^D} = D_{\mathfrak{q}/\mathfrak{p}}$. Similarly, we also get that $I_{\mathfrak{q}/\mathfrak{q}^D} = I_{\mathfrak{q}/\mathfrak{p}}$, which implies $e_{\mathfrak{q}/\mathfrak{q}^D} = e_{\mathfrak{q}/\mathfrak{p}}$. So by multiplicativity, we get that $e_{\mathfrak{q}^I/\mathfrak{q}^D} = 1$.

Note that the above argument also gives us that $f_{\mathfrak{q}/\mathfrak{q}^I} = 1$ (as in general $e_{\mathfrak{q}/\mathfrak{q}^I} f_{\mathfrak{q}/\mathfrak{q}^I} g_{\mathfrak{q}/\mathfrak{q}^I} = [E : E^I]$).

[1] This all just follows from the fundamental theorem of Galois theory. Recall that we have an inclusion-reversing bijection between subgroups H of $\mathrm{Gal}(E/K)$ and intermediate fields of E/K, where H maps to its fixed field E^H.

Now using that $|D_{\mathfrak{q}/\mathfrak{q}^D}| = |\mathrm{Gal}(E/E^D)|$, we have

$$e_{\mathfrak{q}/\mathfrak{q}^I} e_{\mathfrak{q}^I/\mathfrak{q}^D} f_{\mathfrak{q}/\mathfrak{q}^I} f_{\mathfrak{q}^I/\mathfrak{q}^D} = e_{\mathfrak{q}/\mathfrak{q}^D} f_{\mathfrak{q}/\mathfrak{q}^D} = |D_{\mathfrak{q}/\mathfrak{p}^D}| = |\mathrm{Gal}(E/E^D)| = [E : E^D]$$

and thus, as $e_{\mathfrak{q}/\mathfrak{q}^I} = [E : E^I]$, $f_{\mathfrak{q}/\mathfrak{q}^I} = 1$, and $e_{\mathfrak{q}^I/\mathfrak{q}^D} = 1$, this implies

$$f_{\mathfrak{q}^I/\mathfrak{q}^D} = [E^I : E^D]$$

which proves the claim.

(c) Recall that we have the equation

$$\sum_{\substack{\mathfrak{q}' \lhd O_E \,:\, \mathfrak{q}' | \mathfrak{q}^D \\ (\mathfrak{q}' \text{ lies above } \mathfrak{q}^D)}} e_{\mathfrak{q}'/\mathfrak{q}^D} f_{\mathfrak{q}'/\mathfrak{q}^D} = [E : E^D].$$

However, as we already have $e_{\mathfrak{q}/\mathfrak{q}^I} = [E : E^I]$, this implies $e_{\mathfrak{q}/\mathfrak{q}^D} f_{fq/\mathfrak{q}^D} = [E : E^D]$, and thus the above sum can only contain one term. Therefore $\mathfrak{q}$ is the unique prime of E lying above $\mathfrak{q}^D$.

2. Recall that, for any intermediate field L, we have $e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_L} e_{\mathfrak{q}_L/\mathfrak{p}}$ and $f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{q}_L} f_{\mathfrak{q}_L/\mathfrak{p}}$, which follows from multiplicativity in towers of fields.

(a) If $e_{\mathfrak{q}_L/\mathfrak{p}} = 1$ and $f_{\mathfrak{q}_L/\mathfrak{p}} = 1$, then this implies $e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_L}$ and $f_{\mathfrak{q}/\mathfrak{p}} = f_{\mathfrak{q}/\mathfrak{q}_L}$. Therefore, we have that

$$|D_{\mathfrak{q}/\mathfrak{p}}| = e_{\mathfrak{q}/\mathfrak{p}} f_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_L} f_{\mathfrak{q}/\mathfrak{q}_L} = |D_{\mathfrak{q}/\mathfrak{q}^L}|$$

But since $D_{\mathfrak{q}/\mathfrak{q}_L} \subset D_{\mathfrak{q}/\mathfrak{p}}$, this implies $D_{\mathfrak{q}/\mathfrak{q}_L} = D_{\mathfrak{q}/\mathfrak{p}} = \mathrm{Gal}(E/E^D)$. (see exercise 1(a)). In particular, we must have $D_{\mathfrak{q}/\mathfrak{p}} \subset \mathrm{Gal}(E/L)$, and thus by Galois theory we have $L \subset E^D$.

(b) This follows by the same argument as (a), but just using I instead of D. If $e_{\mathfrak{q}_L/\mathfrak{p}} = 1$, then this implies $e_{\mathfrak{q}/\mathfrak{p}} = e_{\mathfrak{q}/\mathfrak{q}_L}$, and thus $|I_{\mathfrak{q}/\mathfrak{p}}| = |I_{\mathfrak{q}/\mathfrak{q}_L}|$. As $I_{\mathfrak{q}/\mathfrak{p}_L} \subset I_{\mathfrak{q}/\mathfrak{p}}$, this implies $I_{\mathfrak{q}/\mathfrak{p}_L} = I_{\mathfrak{q}/\mathfrak{p}} = \mathrm{Gal}(E/E^I)$. Therefore $L \subset E^I$, again by Galois theory.

(c) If $\mathfrak{q}$ is the only prime of O_E lying above $\mathfrak{q}_L$, then $g_{\mathfrak{q}/\mathfrak{q}_L} = 1$. Therefore $|D_{\mathfrak{q}/\mathfrak{q}_L}| = [E : E_L]$, which implies $D_{\mathfrak{q}/\mathfrak{q}_L} = \mathrm{Gal}(E/L)$ and $H \subset D_{\mathfrak{q}/\mathfrak{p}}$. Therefore $E^D \subset L$, again by Galois theory.

3. Let E be the Galois closure of $(L_1 \cdot L_2)/K$. Let $\mathfrak{q}$ denote a prime of O_E lying above $\mathfrak{p}$. We can now apply the previous exercise.

(a) As the prime $\mathfrak{p}$ is unramified in L_1/K and L_2/K, this implies that $e_{\mathfrak{q}_{L_1}/\mathfrak{p}} = 1$ and $e_{\mathfrak{q}_{L_2}/\mathfrak{p}} = 1$, and thus by the previous exercise 2(b) we have that $L_1 \subset E^I$, and $L_2 \subset E^I$, thus $L_1 \cdot L_2 \subset E^I$.

By exercise 1(a), we have that $e_{\mathfrak{q}/\mathfrak{q}_L} = e_{\mathfrak{q}/\mathfrak{p}}$, and thus $e_{\mathfrak{q}^I/\mathfrak{p}} = 1$ by multiplicativity in towers. Thus $\mathfrak{p}$ is unramified in E^I and thus unramified in $L_1 \cdot L_2$ (again by multiplicativity of e)

(b) Note that if $\mathfrak{p}$ splits completely, then this implies that $e_{\mathfrak{q}_{L_1}/\mathfrak{p}} = f_{\mathfrak{q}_{L_1}/\mathfrak{p}} = 1$ and $e_{\mathfrak{q}_{L_2}/\mathfrak{p}} = f_{\mathfrak{q}_{L_2}/\mathfrak{p}} = 1$. Thus a similar argument to part (a) (now using exercise 2(a)) thus proves that $L_1 \cdot L_2 \subset E^D$.

Again my multiplicativity in towers, this gives us that $\mathfrak{p}$ splits completely in E^D and thus splits completely in $L_1 \cdot L_2$.

4. See Neukirch's textbook *Algebraic number theory*, pages 48–49.